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 - Determine which of these 8 are the likely cause
 - Determine the evolutionary origin of the new virus

Methods

- Your tasks:
 - Determine which of these 8 are the likely cause
 - Determine the evolutionary origin of the new virus
- How?
 - Search for homologous sequences in a database of sequenced viral genomes
 - Build a phylogenetic tree of related sequences

BLAST (Basic Local Alignment Search Tool)

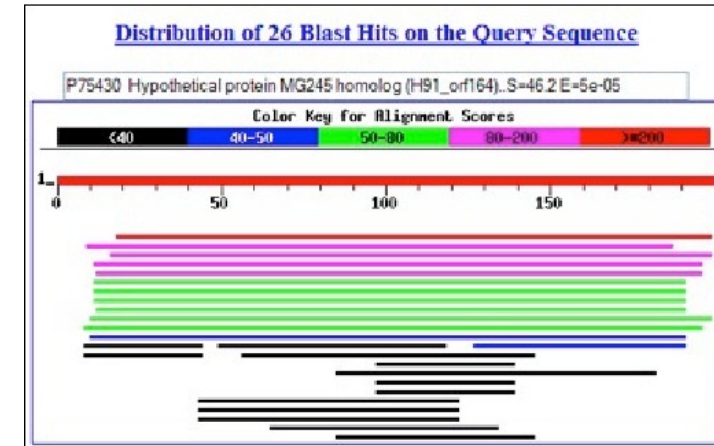
QUERY sequence(s)

```
>gi|15237380|ref|NP_197163.1| myb family transcription factor (MYB43) [Arabidopsis thaliana]
MGRQPCQDKVGLKKGPWTEEDKKLINFILTNMCCWRALPKLSGLLRGGKSCRLRWINYLRPOLKRGLL
SEYEQKVINLHAQLGNRWKXIASHLPGRTDNEIKHWNTHIKKKLRKMGIDPLTHKPLSEQEASQQAQGS
RKKSLVPHDDKNPKQDQQTRODQEQHLEQALEKNNTSVSGDGFCDVPLINPHEILIDISSHHHSN
DONVNINTSKFTSPSSSSSTSSCISVVVPGDEFKFFDEMEILDKWLSSDDSLGDDISKDGKFNNTV
DTMNLWDINDLSSLDMMFMNEHDDGFIGNGNGSRMVLDDQSWTFDLL
```

BLAST program

BLAST
database

BLAST results



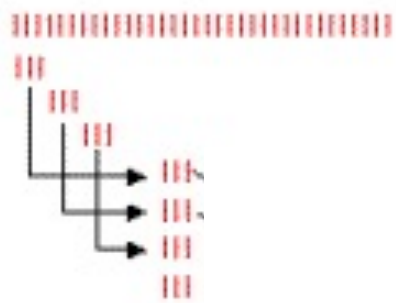
- Search for similarity to infer “homology”

BLAST

- BLAST is optimized to search large databases quickly.
- How does it do this?

BLAST: Heuristic algorithm

Query sequence of length L (this is the sequence with which you do a search)

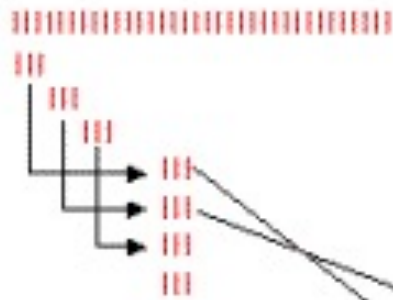


Compile list of words (w) from query
usually $w=3$ for proteins and 11-28 for nucleotides

There are $L-w+1$ words in sequence L
Begin with high scoring words

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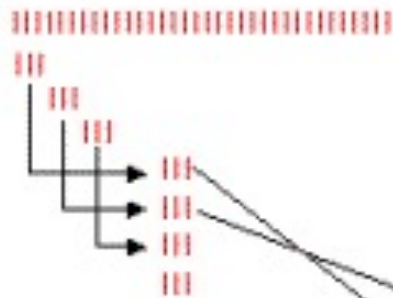
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Compare word list with sequences
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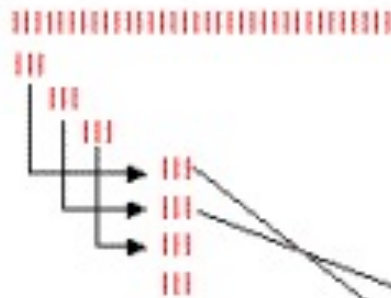
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Extend matches in both directions
until further extension causes the
score to drop by a certain amount

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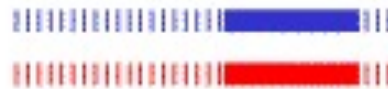
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High scoring segment pair HSP

A scoring matrix is used to evaluate matches

A	4																			
R	-1	5																		
N	-2	0	6																	
D	-2	-2	1	6																
C	0	-3	-3	-3	9															
Q	-1	1	0	0	-3	5														
E	-1	0	0	2	-4	2	5													
G	0	-2	0	-1	-3	-2	-2	6												
H	-2	0	1	-1	-3	0	0	-2	8											
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4										
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4									
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5								
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5							
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6						
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7					
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4				
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5			
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11		
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4
	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V

BLOSUM 62 scoring matrix

(positive values are shaded)

Numbers represent the probability of finding that sequence pair in homology sequences

The values for amino acid substitutions were obtained from Henikoff S & Henikoff JG (1992) Amino acid substitutions matrices from protein blocks. *Proc. Natl. Acad. Sci.* **89**: 10915-10919.

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M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5							
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6						
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7					
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4				
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5			
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11		
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4
	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V

S1: W-A-S-P

S2: W-E-S-T

W-W = 11

A-E = -1

S-S = 4

P-T = -1

Total score for
this alignment: 13

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Q : ROBJOEZACANNLIZ

Break this up into 3 letter words

ROB , OBJ , BJO , . . , ZAC , ...ANN , ...NLI , LIZ

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Search sequences S1, S2, etc. in database

Find a match with the word ZAC then extend on both sides until no
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Q : ROBJOEZACANNLIZ

S1 : TOMZOEZACANNLIA

Q : ROBJOEZACANNLIZ

S2 : TOMZOEZACAMYLEA

Q : ROBJOEZACANNLIZ

Break this up into 3 letter words

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Search with high scoring words first for better chance
of high scoring alignments

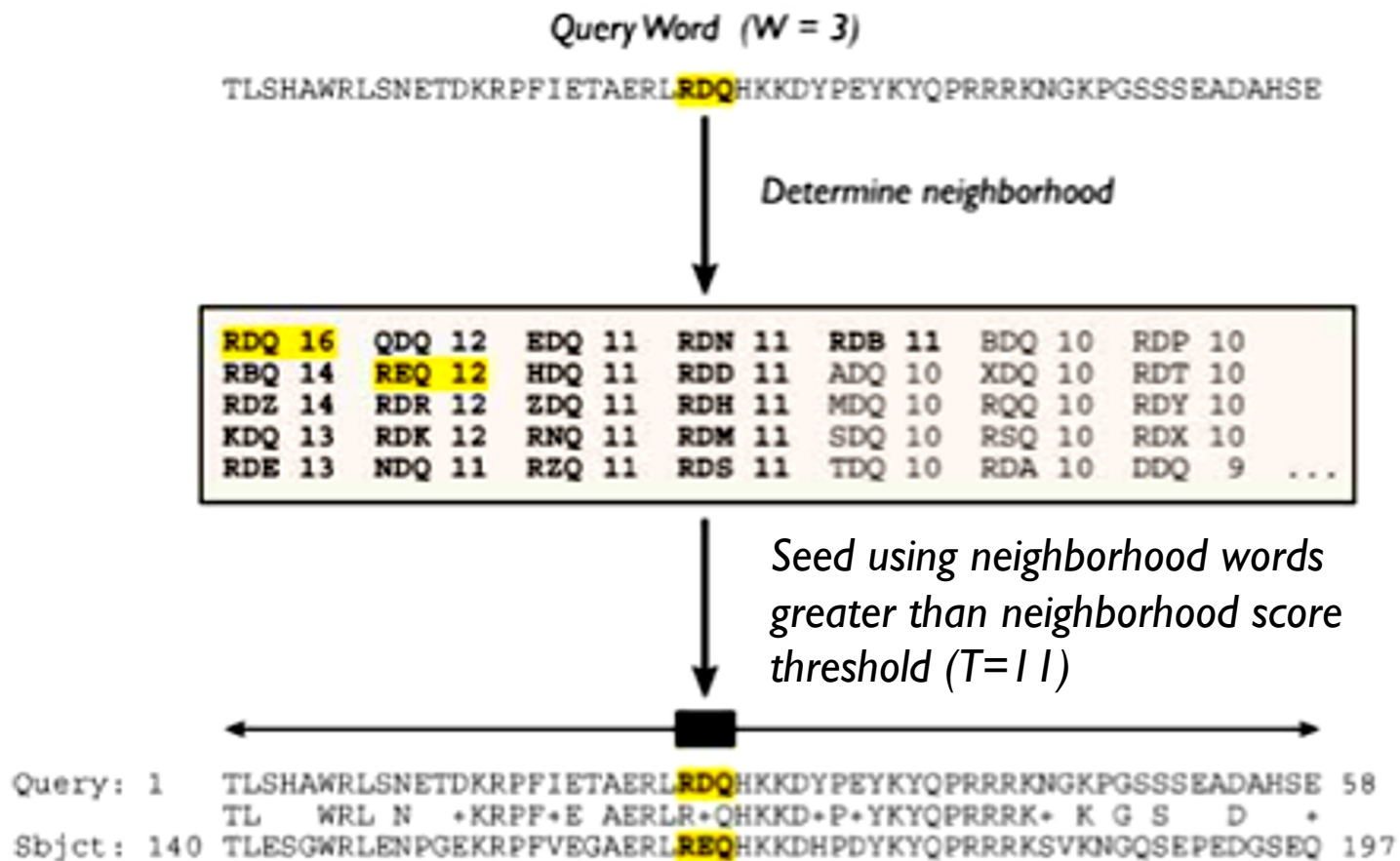
Q : LVA AVGV CWD ILRAAA

In the above example, BLOSUM62 scores for matches to LVA
and CWD are 12 and 26 respectively, so search with CWD

Q : LVA AVGV CWD ILRAAA
 | | | | | | | |
S : AGG AVVV CWD ILKAGG

useful parameters

- Word size: the size of the chunks that the query sequence is chopped into
- Threshold: minimum score for a word match to be considered to seed an extension



How
BLAST
works

HSP = High-scoring Segment Pair – a segment pair whose score will not increase by further extension or by trimming

Score (S) = measures alignment quality (scoring matrix - gaps)

E value (E) = number of different alignments with score S that are expected to occur by chance in a search of that database

Nucleotide vs Protein BLAST

- blastn: nucleotide blast. Comes in different flavors
 - megablast: optimized for nearly identical sequences
 - dc-megablast: discontinuous megablast...more distant sequences

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 - Default word size is 28 (megablast) or 11 (dc-megablast)
 - No threshold for seeding, requires exact match

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- Seeding:
 - Default word size is 28 (megablast) or 11 (dc-megablast)
 - No threshold for seeding, requires exact match
- Scoring matrix
 - Exact match: +1 (megablast); +2 (dc-megablast)
 - Any mismatch: -2 (megablast); -3 (dc-megablast)

BLAST Summary

- Computes regions of high “similarity” in local alignments of 2 sequences
- Break search into “chunks” by finding all subsequences (stretches of similarity, or “words”) of length k that occur in both seqs
- Build score on matches (scoring matrix, gap cost)
- Extend subsequences to see if score increases
- Compute total score (when no more extensions are possible)
- Then compare BLAST score against precomputed expected scores for all sequences in database
- Then rank score

Command Line BLAST

- You are probably familiar with the web interface for BLAST
- We will use a command-line version of the program
- Why would one want to do this?
 - Overcome web version limitations on query size
 - E.g. BLAST one genome against another
 - Can use custom database
 - Easier to test the effect of changing parameters
 - Torture